

K935 — CATCH (flagship, timely — Cal is re-reading now): line 114’s “Qualitative QCD dynamics — DERIVED (K755)” OVER-CLAIMS. The reasoning it gives for “the AF direction is derived” —  $\beta_0 = 11N_c - 2N_f = 21 > 0$  from  $N_c=3$ ,  $N_f=6$  content — is the IMPORT (BST’s integers plugged into the STANDARD  $\beta$ -function formula; the 11 and 2 are standard QFT, not BST-derived). Today’s entire strong-sector discipline (K927 charter → K934) established that this is an IDENTIFICATION, not a derivation. The genuine geometric derivation of the sign (short roots → non-abelian SU(3) →  $E=-2F$  antiscreening, Elie today) is CONDITIONAL PASS (K934) — real but not yet full (needs the one-domain recast). So the flagship’s flat “DERIVED” is ahead of what’s honestly banked. Fix queued for Lyra; flagged to Cal so his re-read factors it in. Do NOT edit mid-re-read (collision/PDF-restale risk) — Lyra applies after Cal’s pass.

Keeper | 2026-07-26 Sun | Today’s sharper import-vs-derive discipline exposed a stale over-claim in the flagship (line 114, citing the older K755). Caught while Cal re-reads — flag the referee, queue the author, don’t edit under him.

The over-claim (verbatim, line 114)

Qualitative QCD dynamics — DERIVED (K755): the sign of the QCD  $\beta$ -function follows from the content —  $\beta_0 = 11 N_c - 2 N_f = 21 > 0$  ( $N_c = 3$ ,  $N_f = 6 = 3 \text{ generations} \times 2$ )  $\Rightarrow$  asymptotically free in the UV, confining in the IR ... The *running value*  $\alpha_s(\text{scale})$  is a runner; the *direction* is derived.

Why it over-reads (today’s discipline)

- “The sign follows from the content” = BST supplies  $N_c=3$ ,  $N_f=6 \rightarrow$  plug into the STANDARD formula  $11N_c-2N_f \rightarrow \text{sign}>0$ . The 11 and 2 are standard-QFT loop coefficients, NOT BST-derived. This is precisely the “asymptotic freedom using standard coefficients  $\neq$  asymptotic freedom derived” line drawn in K927/K929/Grace’s book-source. Labeling it flatly “DERIVED” is the exact conflation the session’s FF-20/import-vs-derive discipline exists to prevent.
- The genuine geometric derivation of the sign landed TODAY (Elie, K934): short-root count  $\rightarrow$  non-abelian SU(3) (T666)  $\rightarrow$  the gluon’s  $E=-2F$  paramagnetic antiscreening. That is target-innocent and matched two blind bars — but it is CONDITIONAL PASS, pending the one-domain recast (K933). So even the honest “derived” is not yet full.
- Net: line 114 asserts as banked (“DERIVED”) what is (a) an identification via the imported formula, plus (b) a conditional-pass geometric result. Over-claim.

The FIX (queued for Lyra — precise wording)

Replace “Qualitative QCD dynamics — DERIVED (K755)” with the honest two-part tier:  $>$  Qualitative QCD dynamics — the AF direction. Given the *standard*  $\beta$ -function structure, BST’s content ( $N_c=3$ ,  $N_f=6=3 \text{ gen} \times 2$ ) fixes the sign:  $\beta_0 = 11N_c - 2N_f > 0 \Rightarrow$  asymptotically free in the UV, confining in the IR — consistent with Schur-confinement (§3). This is identification-level (the 11/3, 2/3 loop coefficients are standard, not BST-derived). A target-innocent geometric derivation of the sign — the geometry forcing non-abelian SU(3) (short roots, T666), whose gluon color-magnetic moment ( $E=-2F$ ) antiscreens — is emerging (K934, conditional on a one-domain recast). The running *value*  $\alpha_s(\text{scale})$  is a runner. This is STRONGER, not weaker: it cites the new geometric result as the path to a real derivation while being honest that the content-plug is an identification.

Also (minor, same fix pass)

- Line 114 writes  $\beta_0 = 21$  (un-normalized  $11N_c - 2N_f$ ); today's work used the  $/3$ -normalized  $\beta_0 = 7 = g$  (and flagged  $7=g$  as an identification-coincidence, not a target). Convention is fine; just don't let "21" or " $7=g$ " read as a BST derivation of the coefficient.
- Confirm line 117 ("the running direction  $\alpha_s$  asymptotically free, §9") is consistent with the softened 114 — it's fine ("direction" from content), but keep the identification/derivation tiering aligned.

## Disposition

- Cal: factor this into the re-read — line 114 is a known Keeper-caught over-claim; the corrected wording is above. (Your re-read need not block on it; the fix is queued.)
- Lyra: apply the fix after Cal's pass (avoid editing under his re-read); rebuild PDF.
- No partition-theorem impact — this is the strong-sector qualitative-dynamics line (§Frontier/§9-adjacent), not the capstone. But it IS in the flagship headed for external, so it must be honest before it goes.

— Keeper K935, 2026-07-26. CATCH: flagship line 114 "Qualitative QCD dynamics — DERIVED (K755)" over-claims — the  $\beta_0=11N_c-2N_f=21>0$ -from-content reasoning is the IMPORT (standard formula at BST content; 11,2 not BST-derived) = an IDENTIFICATION per today's discipline (K927/K929/Grace). The genuine geometric sign-derivation (Elie K934: short roots→non-abelian  $SU(3)$ → $E=-2F$  antiscreening) is CONDITIONAL PASS, not full. Fix queued (Lyra): soften to "identification-level from content + target-innocent geometric derivation emerging (K934, conditional)" — stronger and honest. Flagged to Cal (re-reading now); Lyra applies after his pass (no mid-re-read edit). No partition-theorem impact but it's flagship→external, must be honest. See [[Keeper\_K934\_AUDIT\_elie\_a2\_AF\_sign\_CONDITIONAL\_PASS]], [[Keeper\_K927\_STRONG\_SECTOR\_DYNAMICS\_area\_chapter]], [[feedback\_qft\_textbook\_framing]].